

How Weather and Temperature Change Daily Bike Rentals: Evidence from Capital Bikeshare in Washington, DC

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Introduction

Bike sharing is a cheap, low-emission way to get around a city, but it only works if bikes are available when people want them. Because cycling happens outdoors, the number of rentals on a given day should depend on the weather. A bike-share company can use this to plan how many bikes to put out.

In this report, I address a simple question: **how do weather and temperature change the number of bikes rented per day?** I split it in two parts. First, how many more rentals are there for every extra degree Celsius? Second, how many fewer rentals are there on rainy days than on clear days?

To answer this, I analyse the public Bike Sharing Dataset of Capital Bikeshare in Washington, DC (USA), hosted by the UCI Machine Learning Repository [1]. All analysis was done in Python with numpy and matplotlib.

Data sample

The dataset contains one row for each of 731 days, from 1 January 2011 to 31 December 2012 [1]. The columns used in this analysis are:

- `yr`: year (0 for 2011, 1 for 2012)
- `weathersit`: weather type (1 clear or partly cloudy, 2 mist or cloudy, 3 light rain or snow)
- `temp`: temperature, stored divided by 41
- `cnt`: total number of rentals on that day

The dataset needed little cleaning: there were no missing values, and I only had to multiply `temp` by 41 to get degrees Celsius. Rainy days are rare: only 21 of the 731 days have weather type 3, and only 6 of them are in 2012.

Method

For each group of days I computed the mean number of rentals $\bar{N}$. As statistical uncertainty I used the sample standard deviation σ (`np.std` with `ddof=1`). It describes how much a single day differs from the mean, and it does not get smaller when more days are added. For two independent groups, the uncertainty of the difference is $\sigma_{\text{diff}} = \sqrt{\sigma_1^2 + \sigma_2^2}$. In formulas:

$$\bar{N} = \frac{1}{n}\sum_{i=1}^{n} N_i, \qquad \sigma = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}\left(N_i - \bar{N}\right)^2}, \qquad \hat{N} = aT + b.$$

Here N_i are the daily rentals, T the temperature, a the slope and b the intercept.

To study temperature, I averaged the rentals in bins of 5 °C (bins with fewer than 5 days were skipped), fitted a straight line through all days with `np.polyfit`, and computed the Pearson correlation r with `np.corrcoef`. The uncertainty on the slope was estimated with the sub-sampling technique: I shuffled the days (fixed seed 42), split them into 10 groups of about 73 days, fitted a slope in each group, and took the mean and standard deviation of the 10 slopes. Because rentals grew a lot from 2011 to 2012, I finally repeated the analysis for each year separately.

Results and discussion

I first looked at the weather. Figure 1 (right) shows the mean number of rentals per day for each weather type. Clear days have the most rentals and rainy or snowy days the fewest:

$$\bar{N}_{\text{clear}} = 4877 \pm 1880, \qquad \bar{N}_{\text{mist}} = 4036 \pm 1809, \qquad \bar{N}_{\text{rain/snow}} = 1803 \pm 1240.$$

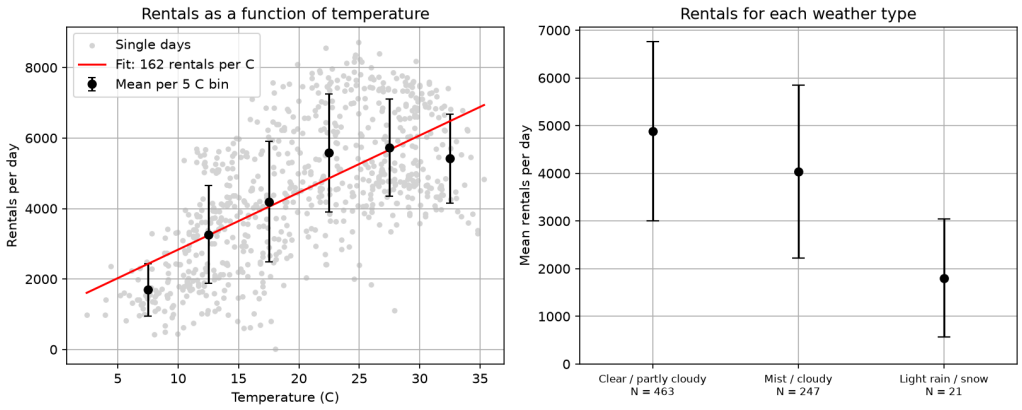


Figure 1: **Left:** rentals as a function of temperature. Grey dots are single days, black points are the mean per 5 °C bin, the red line is the straight-line fit. **Right:** mean rentals per day for each weather type; N is the number of days. Error bars are standard deviations.

The difference between clear and rainy days is

$$\Delta N = \bar{N}_{\text{clear}} - \bar{N}_{\text{rain/snow}} = 3074 \pm 2252,$$

so there are about 63 % fewer rentals on rainy days. This difference is larger than the day-to-day spread within either group (1880 and 1240), but the quoted ±2252 describes how much single days vary, not how precisely the means are known, so I do not read it as a significance test.

Next, I looked at temperature. Figure 1 (left) shows that rentals rise with temperature, from about 1700 per day at 5–10 °C to about 5700 at 25–30 °C. The straight-line fit gives a slope of 162 rentals per day per °C with a correlation of $r = 0.63$, and the sub-sampling gives

$$\text{slope} = 160 \pm 24 \text{ rentals per day per } ^\circ\text{C}.$$

Every extra degree therefore goes together with roughly 160 extra rentals. The relation is not perfectly linear, though: above about 25 °C the bin means level off (5726 at 25–30 °C and 5418 at 30–35 °C), so the straight line overshoots on hot days.

Finally, the year-by-year check. The mean number of rentals grew from 3406 ± 1379 in 2011 to 5600 ± 1789 in 2012, so the system grew a lot. Still, the temperature slope is similar in both years (137 in 2011 and 177 in 2012, with $r = 0.77$ and $r = 0.71$), and rainy days have fewer rentals in both (55 % fewer in 2011, 65 % in 2012). The 2012 rain result rests on only 6 days and should not be over-interpreted. The growth between the years therefore does not explain the weather and temperature effects away.

These results show an *association*, not proof of cause and effect: temperature is tied to the season, and weekdays and holidays also play a role. In short, warmer and clearer days in Washington come with clearly more bike rentals, by about 160 per extra degree and about 63 % fewer on rainy days.

References

[1] H. Fanaee-T. *Bike Sharing Dataset.* UCI Machine Learning Repository (2013). Available at: `https://archive.ics.uci.edu/dataset/275/bike+sharing+dataset` (accessed October 2026).